

Health IT Diffusion and Physician Density

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Rising US HIT investments enables automation

- The US allocated over \$30 billion to health IT adoption (e.g., EHR)
 - Health Information Technology for Economic and Clinical Health (HITECH) Act and indirect provisions in Affordable Care Act (ACA)
- Extensive research on HIT impact on quality and cost
- But: limited understanding of **HIT** impact on **physician density**
 - Focus: **Computerized Provider Order Entry (CPOE)**, an advanced HIT and the key component of HITECH Meaningful Use (MU) criteria
 - E-enter treatment orders, integrate patient info and clinical guidelines, and communicate with pharmacy, laboratory, and other departments
 - In contrast, basic HIT systems, e.g., clinical decision systems, store patient data in digital database and provide clinical treatment support
- Research question: How does **advanced HIT adoption** affect **aggregate hospital-based (HB) physician density** at the county level?
 - # physicians **per 100k** population, **financial** measures, patient **care volume**

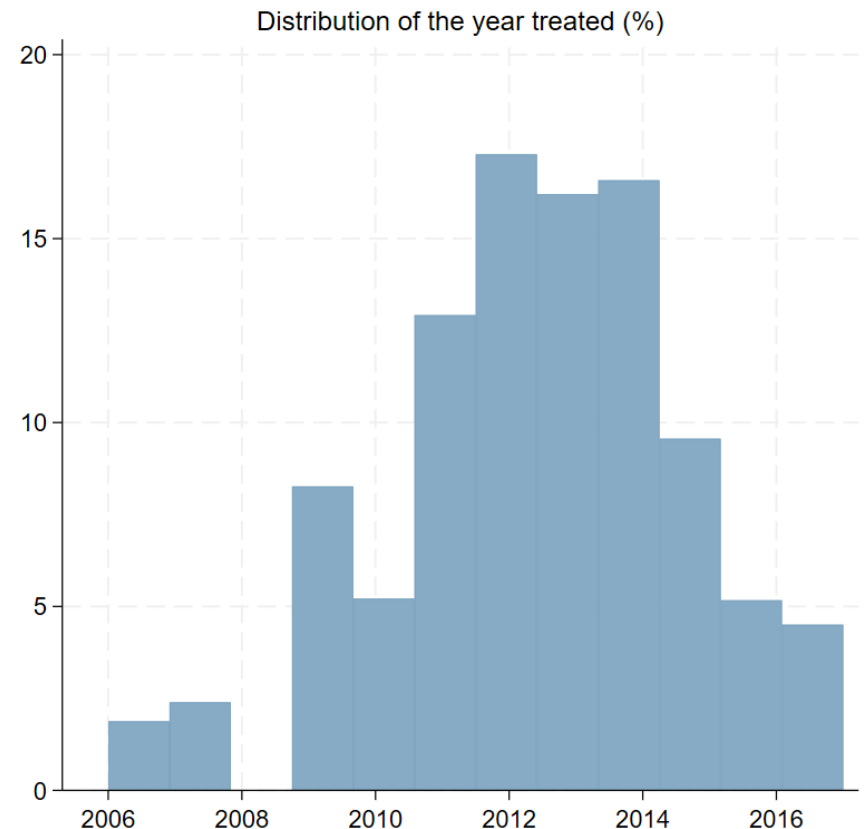
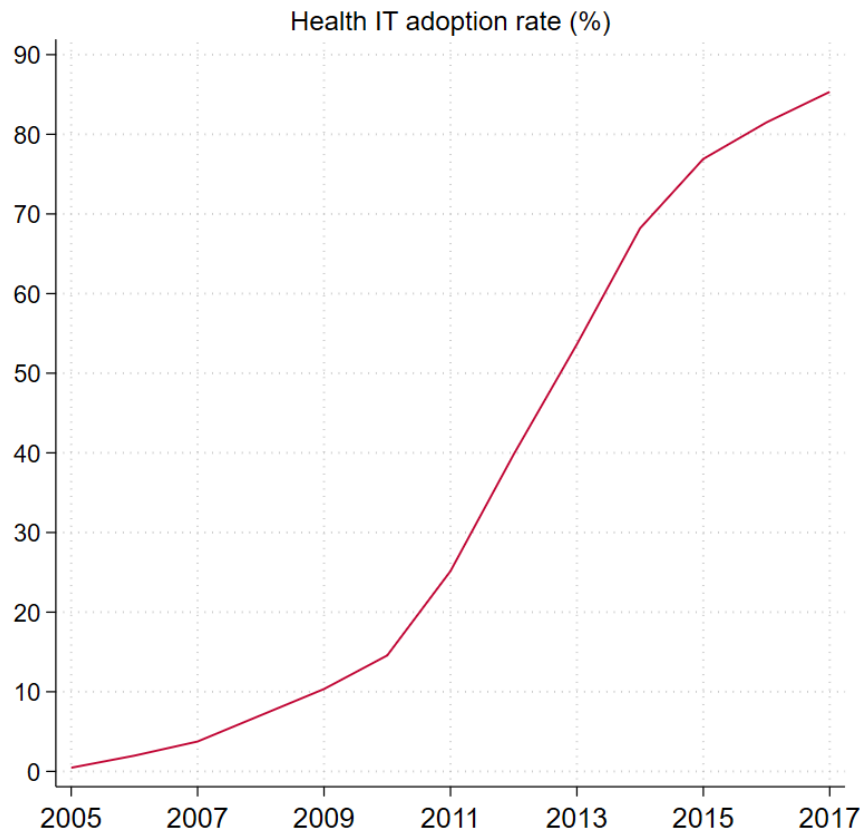
Research Question:

Q: How does advanced HIT adoption affect aggregate hospital-based physician density at the county level?

- Intuitively, aggregate physician rate can go either way (empirical Qs):
 - More: “Time to hire more physicians to take full advantage of HIT!”
 - Less: “Time to automate and reduce docs (recover costs from HIT)!”
 - Same: “Docs come and go, on the net things may stay similar?”
- Exploit the county-level sharp adoption of advanced HIT (CPOE)
- Show how the plausibly exogenous HIT diffusion patterns affect county-level HB physician per 100k pop, the heterogeneity by specialties, physician career stage, physician shortage area status
- Test financial mechanism and downstream impact on care utilization

Institutional context and HIT adoption trends

- Electronic health records (EHRs), the core component of HIT, were invented in 1970s but saw limited adoption until the 2000s. We focus on hospital HIT.
- 2010s: widespread hospital HIT adoption, driven by federal incentives
 - 2009 HITECH Act provides incentive payments for HIT meaningful use (MU); ACA indirect incentives (initiatives/provisions) require/facilitate HIT use.



HIT adoption measures: CPOE & its functionality

- Computerized Provider Order Entry (CPOE): advanced HIT, [automation](#)
 - Enable providers to [electronically enter](#) medication, laboratory, and radiology orders, often with built-in [clinical decision support](#) tools, core in [MU criteria](#)
 - Unlike other HIT component, CPOE directly [structures patient data](#) to [align with coding and billing requirements](#), esp. relevant when payment matters

Order Entry

Alarms ADT Layout HIPAA Help Close

Facility: ADL Goodsam Demo Unit: 01 Resident: CARSON, JOHN

Acct: 123510 MedRecNo: 123510 Room: Admit: 11-02-03 Disch: Male Age: 95 Wgt: 149 lbs PCP:

Rx	End Date	Status	Procedure	End Date	Status	Lab Type	End Date	Status	Rehab Type	End Date	Status
ASPIR-TRIN 325	11-10-07	ACTIVE	APPLY OINTMEI	11-10-07	ACTIVE	BLOOD	10-07-07	DRAFT	PHYSICAL THEF	10-04-07	D/C
COLUMADIN 2.5	11-10-07	ACTIVE	CHANGE DRESS	11-10-07	DRAFT	URINE	09-30-07	ACTIVE			
LASIX 20 MG	11-10-07	ACTIVE	CHANGE DRESS	10-07-07	DRAFT						
ENEMA READY	11-04-07	ACTIVE	O2@2LM NC	09-30-07	ACTIVE						
BISACODYL 10	11-04-07	ACTIVE	APPLY OINTMEI	09-30-07	ACTIVE						
EGL MILK OF M	11-04-07	ACTIVE	CHANGE DRESS	09-30-07	ACTIVE						
ZYPREXA 10 M	10-26-07	ACTIVE									
ASPIR-TRIN 325	10-06-07	DRAFT									
NORVASC 5 MC	09-30-07	DRAFT									
COLUMADIN 6 M	09-30-07	ACTIVE									
BUSPAR 5 MG	109-30-07	ACTIVE									
FOSAMAX 70 M	09-29-07	ACTIVE									
ZYPREXA ZYDI	09-24-07	D/C									
ZYPREXA ZYDI	09-23-07	D/C									
ZOFIRAN 4 MS	08-16-07	ACTIVE									
SINEMET-25/10	09-16-07	ACTIVE									
LASIX 20 MG	109-06-07	ACTIVE									
ACETAMINOPHE	08-24-07	ACTIVE									
BL ZINC GLUCO	08-24-07	ACTIVE									

LABS

REHAB

DIETS

RADIOLOGY

ACTIVITIES

Date	Dx	Description	Allergies	Precautions	09-10-07	07-28-07	N/A	N/A	N/A
09-04-2007	332.	PARKINSON'S DISEASE*			Temp	99			
09-04-2007	331.0	ALZHEIMER'S DISEASE			Pulse	78			
					Resp	14			
					SBP / DBP	121 / 76	126 / 82		
					Weight	149			

Virtual Body Care Plan Progress Notes ADLs Visits Notifications Summary

Conceptual Considerations

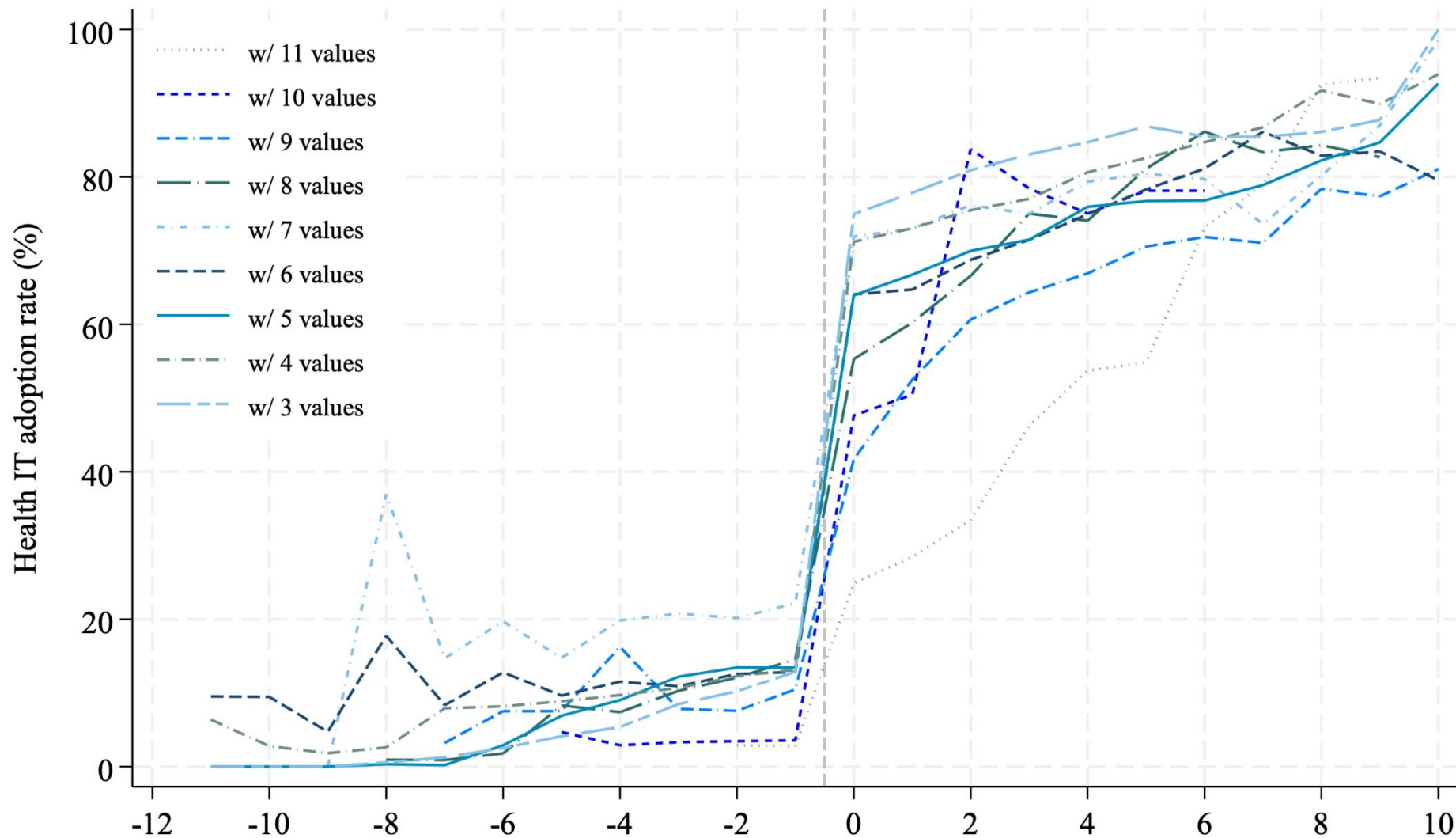
- Local physician density (per 100k pop) is affected by aggregating individual **location choices** and broader **market dynamics**.
- **Hospitals'** perspectives:
 - HIT investment can affect strategic hiring of docs to fully leverage technology
 - Esp in geo-areas with *moderate* but *growing* demand for medical services
 - If increase patient throughput, boost revenue -> can hire/retain more docs
- **Physicians'** perspectives
 - Response to HIT diffusion may vary by specialty, career stage, tech-savviness
 - HIT can be a pull factor for early-career physicians adapted to digital tech
- **Patients'** perspectives
 - HIT can make access to services more convenient with faster scheduling
 - can seek elective procedures earlier and easier (limited sorting within county)

Data Construction

- **Physician supply data:** Area Health Resources Files (AHRF)
 - Hospital-based (HB) physicians: non-federal MDs under hospital contracts providing direct patient care, measured as rate per 100k county population
 - **Specialty categorization:** medical, surgical, general practitioners, others
- **Health IT adoption data:** Healthcare Information and Management Systems Society (HIMSS) Analytics Database
 - Records basic and advanced HIT adoption by hospitals, including CPOE
 - County-level adoption rate weighted by capacity (hospital beds)
- **Medicare provider and service data:** CMS Medicare physician & other practitioners by provider and service dataset (available starting 2013)
- **Market structure and profit data:** Healthcare Provider Cost Reporting Information System (HCRIS), CMS; calculate hospital HHI and profit
- **Sample:** ~2,394 US counties during 2006-2018 outcome years matched with one-year lagged HIT adoption (excl. 2009 b/c missing AHRF data)

HIT treatment measure: sharp jumps in HIT rates

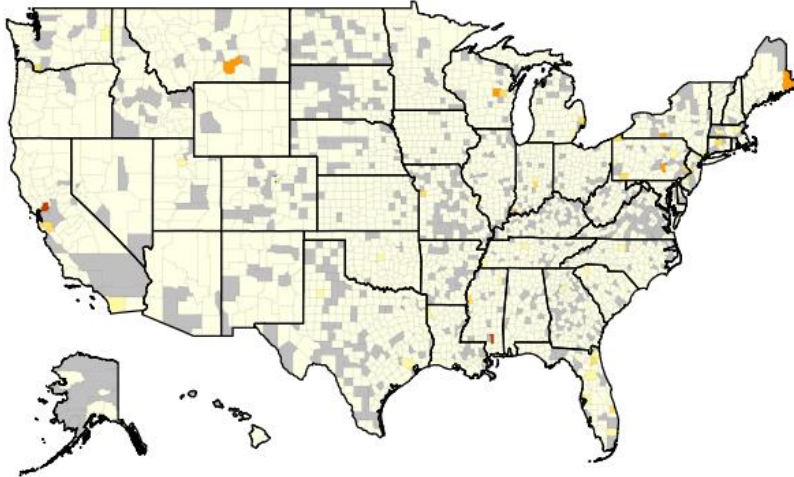
- Define treatment/control group à la the strategy in East et al. (AER 2023)
- Identify the sharpest jump in year τ (roughly = if $0 \rightarrow +$, or $\uparrow 100+\%$ at τ)



Geographic variation in HIT and physician supply

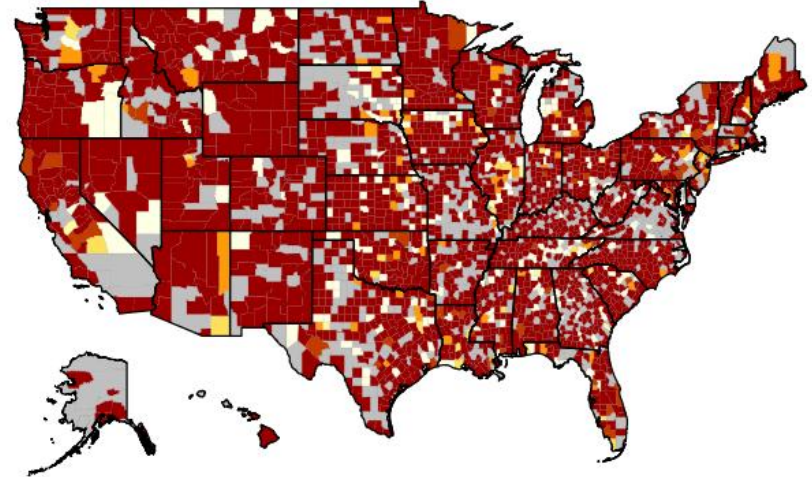
- Outcome periods 2006-2018 with one-year lagged HIT 2005-2017

HIT adoption rate (2005)



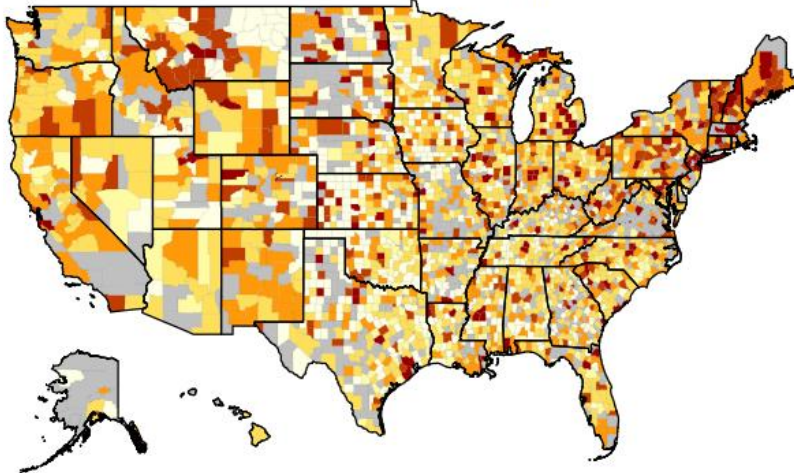
80 <= 100% 60 <= 80% 40 <= 60%

HIT adoption rate (2017)



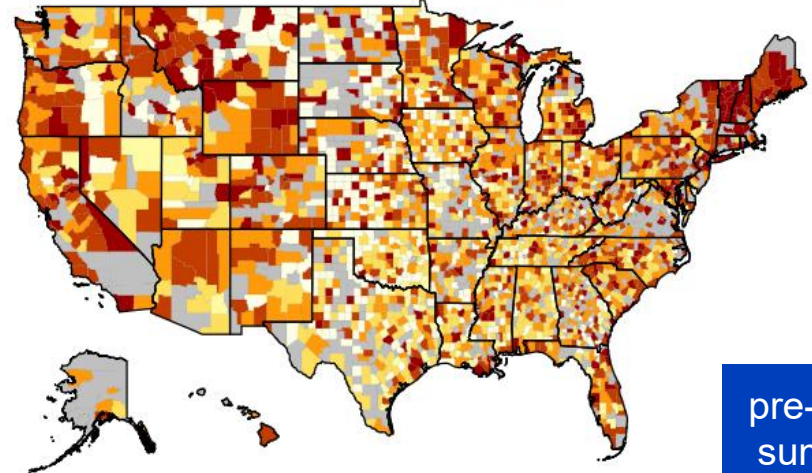
20 <= 40% 0 <= 20% 0 NA

hospital-based physicians per 100k population (2006)



>40.23 21.49 <= 40.23 10.94 <= 21.49

hospital-based physicians per 100k population (2018)



4.07 <= 10.94 0 <= 4.07 0 NA

Empirical strategy: dynamic DiD & event studies

- Staggered DiD design exploits the variation in the timing of the sharp increase in county HIT adoption (De Chaisemartin & d'Haultfoeuille, 2024)

$$Y_{it} = \beta_0 + \beta_1 Treat_{i(t-1)} + \gamma X_{it} + \lambda_{st} + \delta_i + \delta_t + \varepsilon_{it}$$

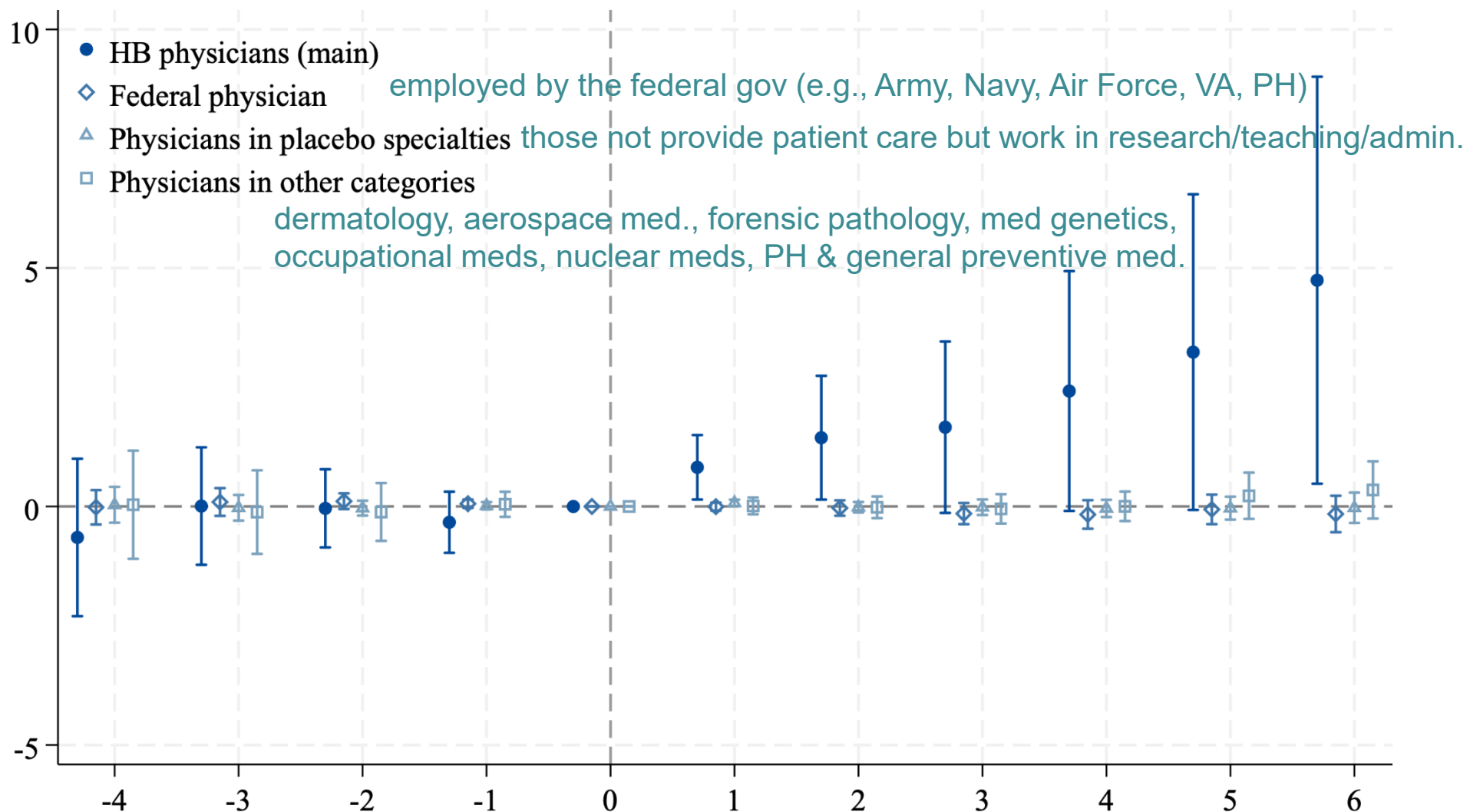
- Y_{it} : physician density in t matched with one-year lagged HIT treatment ($t-1$)
 - X_{it} : 2005 county demographics (share aged 65+, poverty rate, income per capita, Medicare Advantage penetration, share eligible for Medicaid, and college attainment rate), each x a linear time trend to capture time variation while mitigating endogeneity from treatment, à la Dranove et al (2014)
 - λ_{st} : state-specific time trends, where s denotes the state of county i
 - δ_i & δ_t : county and year fixed effects; ε_{it} clustered at the state level
 - Regressions are weighted by county-year level population
- Event study specification:

$$Y_{it} = \beta_0 + \sum_{\tau=-4, \tau \neq 0}^6 \beta_1^\tau \times 1\{t = \tau\} EverTreated_i + \gamma X_{it} + \lambda_{st} + \delta_i + \delta_t + \varepsilon_{it}$$

- Reference period $t=0$, treatment occurred in year 1 (as in dCDH 2024 setup)

HIT diffusion & HB physician: main results

- Main outcome: HB physicians per 100k pop. vs. three falsification groups



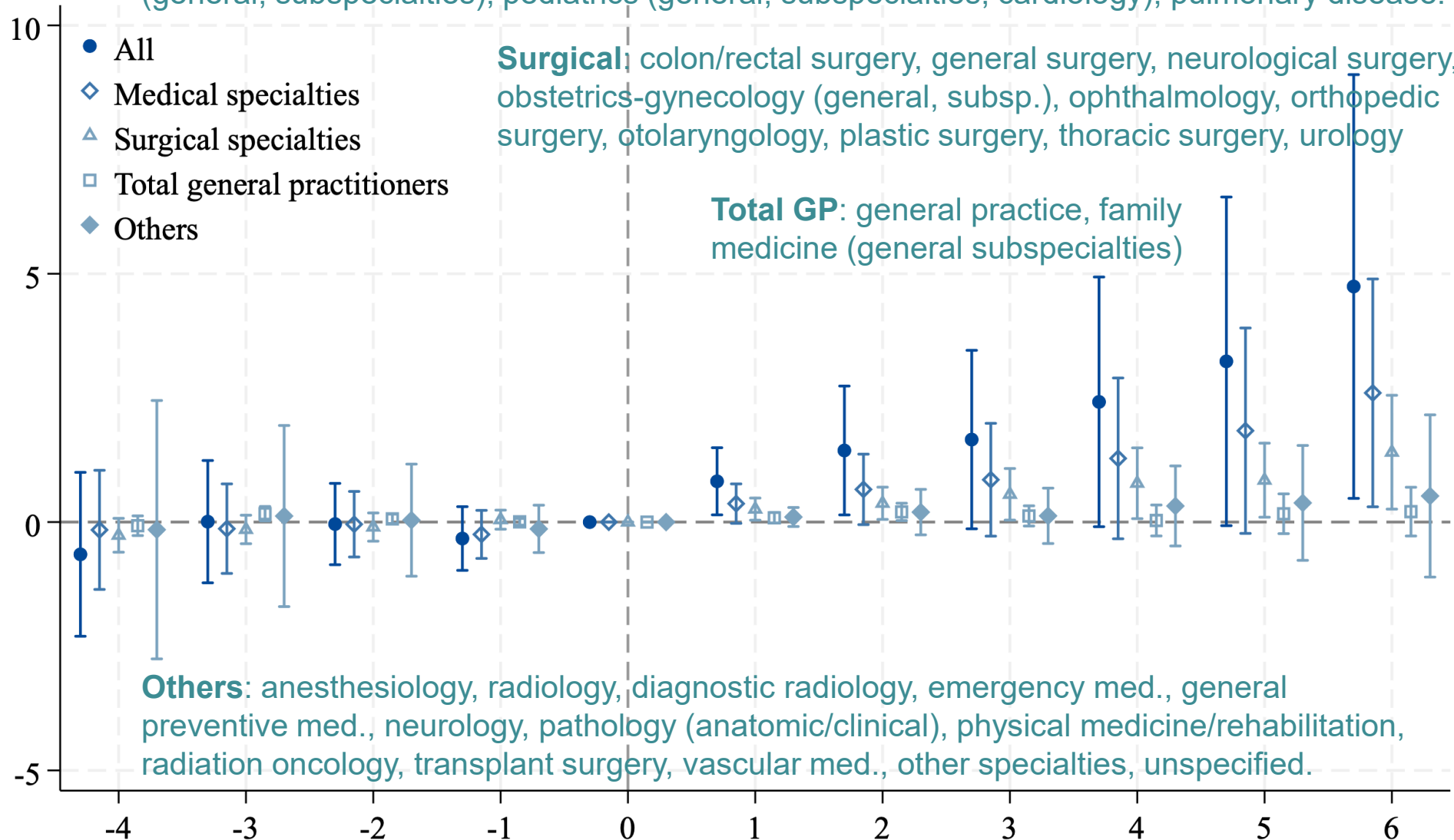
Heterogeneity cross-specialty within HB physicians

Medical: allergy and immunology, cardiovascular disease, gastroenterology, internal medicine (general, subspecialties), pediatrics (general, subspecialties, cardiology), pulmonary disease.

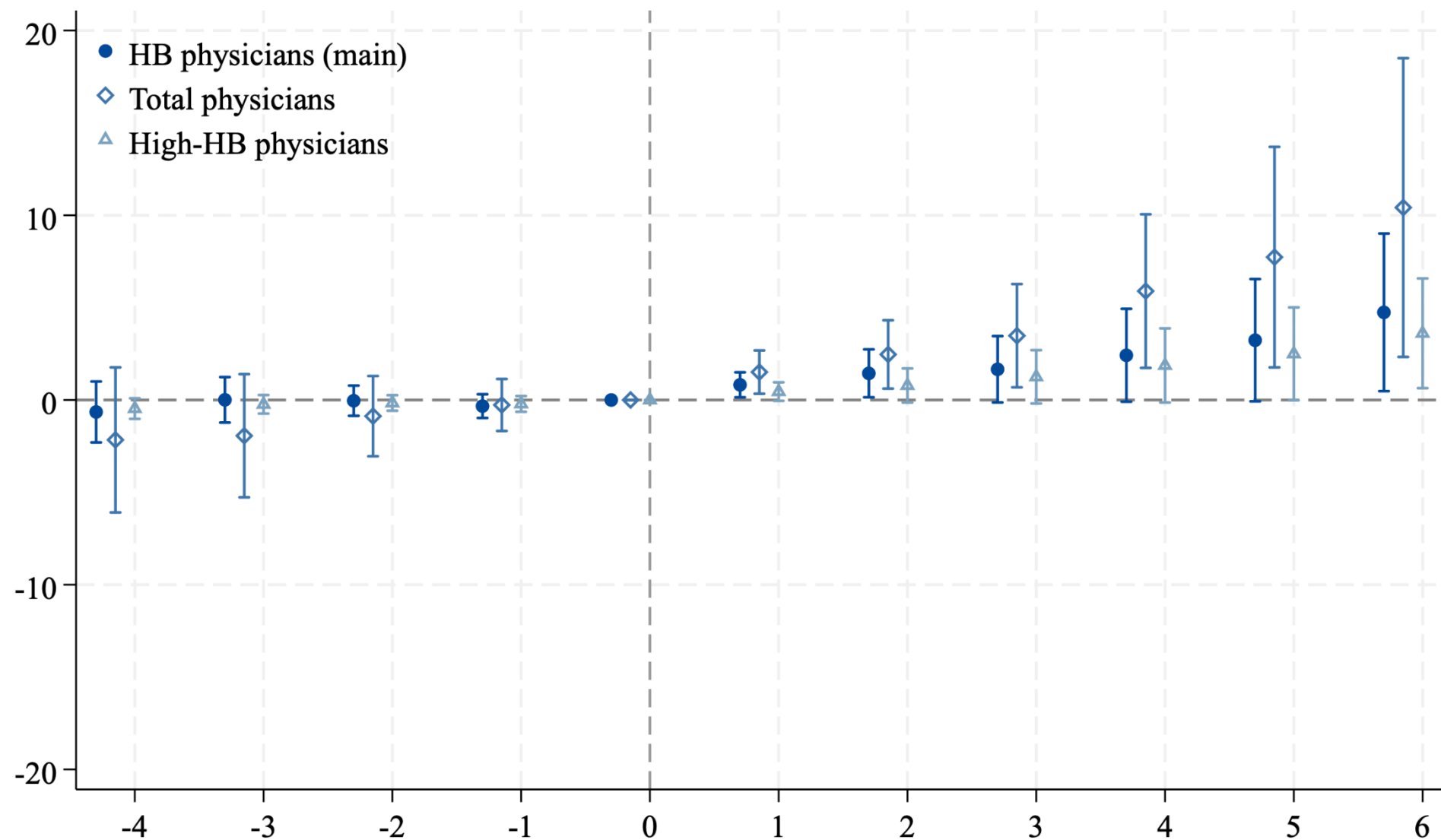
Surgical: colon/rectal surgery, general surgery, neurological surgery, obstetrics-gynecology (general, subsp.), ophthalmology, orthopedic surgery, otolaryngology, plastic surgery, thoracic surgery, urology

Total GP: general practice, family medicine (general subspecialties)

Others: anesthesiology, radiology, diagnostic radiology, emergency med., general preventive med., neurology, pathology (anatomic/clinical), physical medicine/rehabilitation, radiation oncology, transplant surgery, vascular med., other specialties, unspecified.

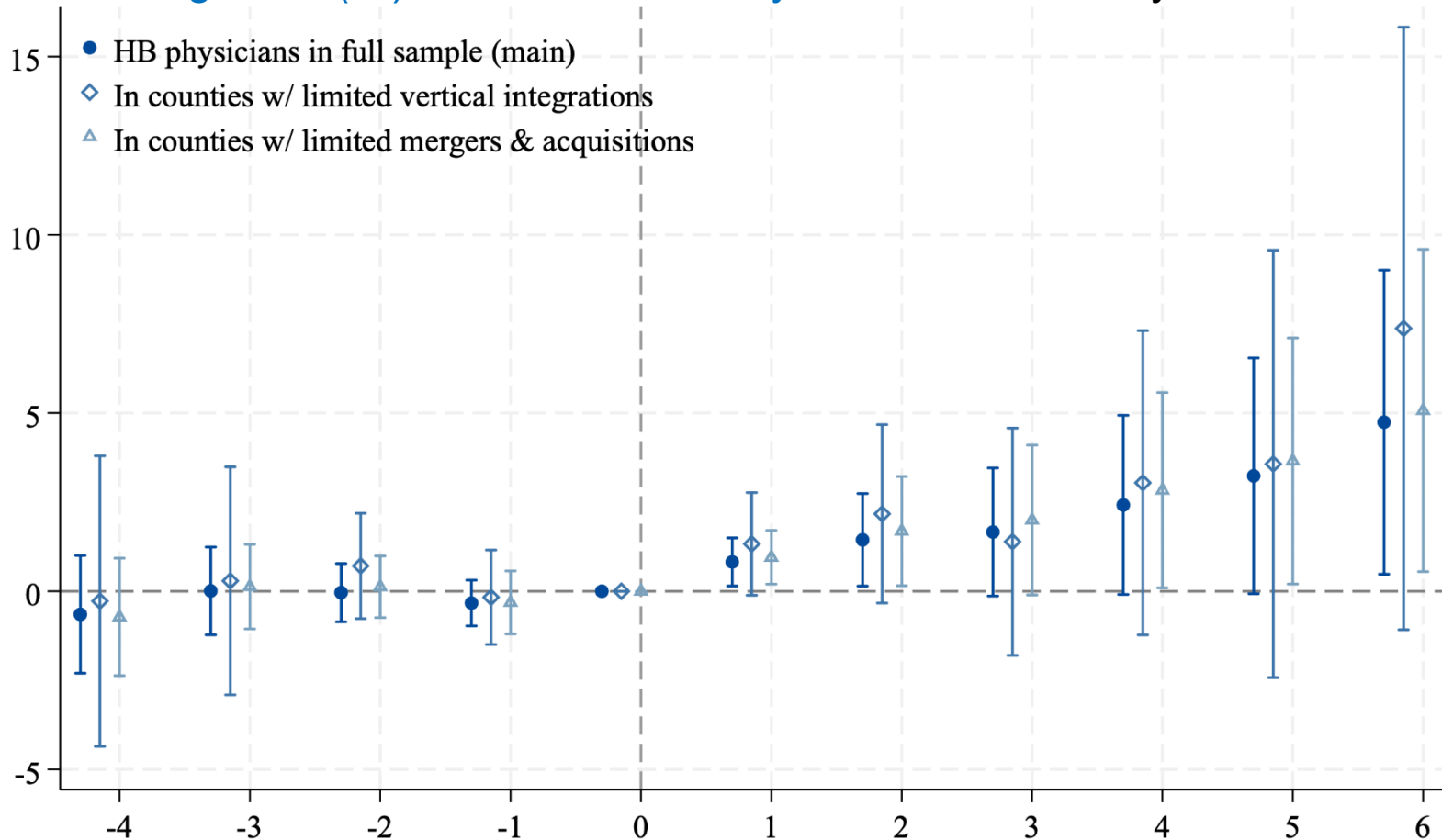


Expanded firm boundary or higher total doc density?



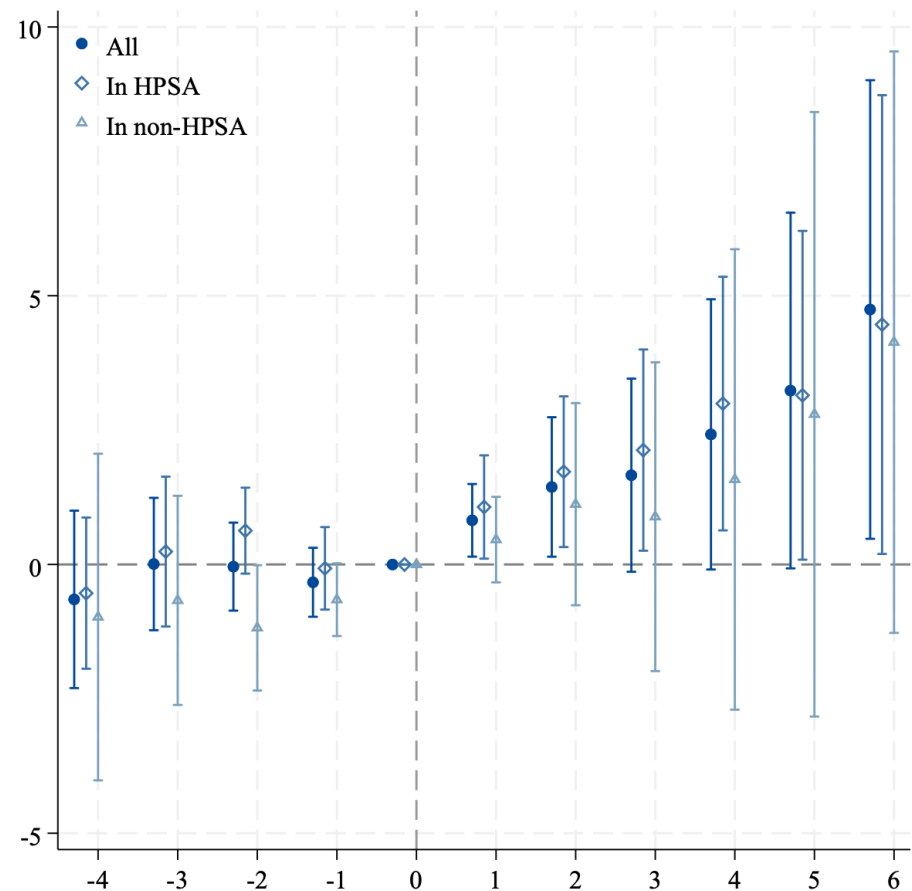
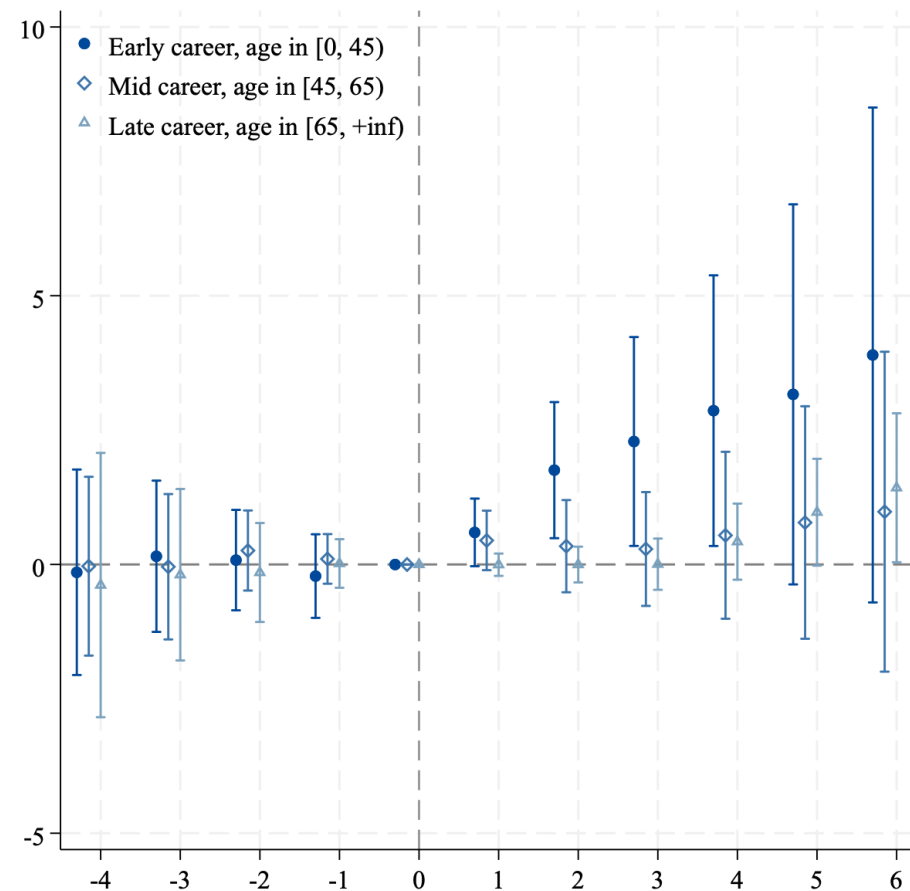
HIT Diffusion or Physician Practice Consolidation?

- Health systems acquire physician practices to get more patients (Baker et al., 2016)
- Systems, w greater financial & tech capacities, often adopt HIT early (Agha 2014)
- We identify health system that experienced a **merger & acquisition (M&A)**, if the **# hospitals within the system increased** in a given year compared to the previous year.
- a **vertical integration (VI)**, if the **# ambulatory care units** in the system increased...



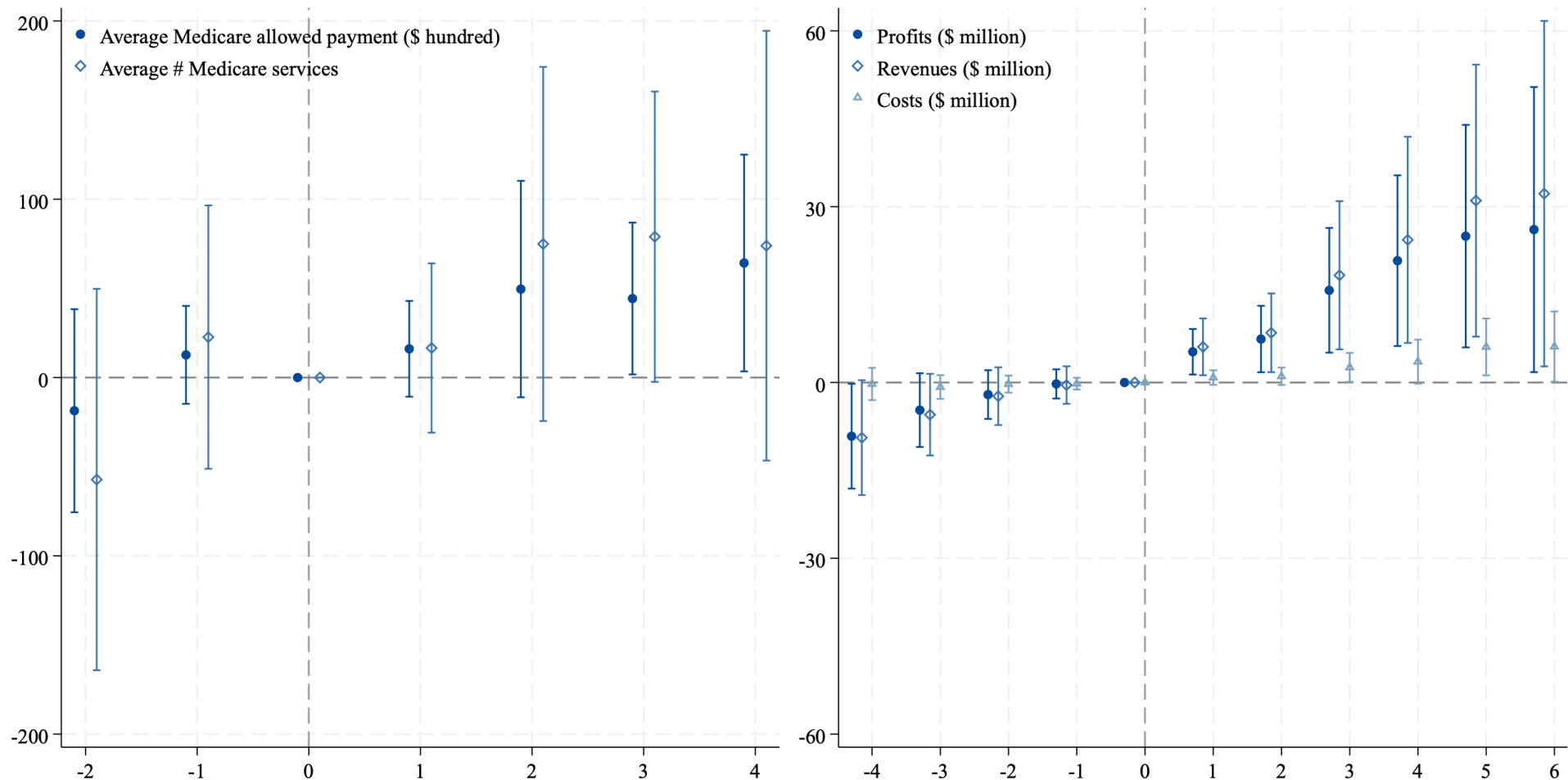
Heterogeneity by Physician Age & Shortage Status

- Early-career (under 45), mid-career (45-64), late-career (65+), total docs
- by Health Professional Shortage Area (HPSA) designation status
 - A county is ever-HPSA if at least part of the county was ever designated as having a shortage of primary care practitioners.



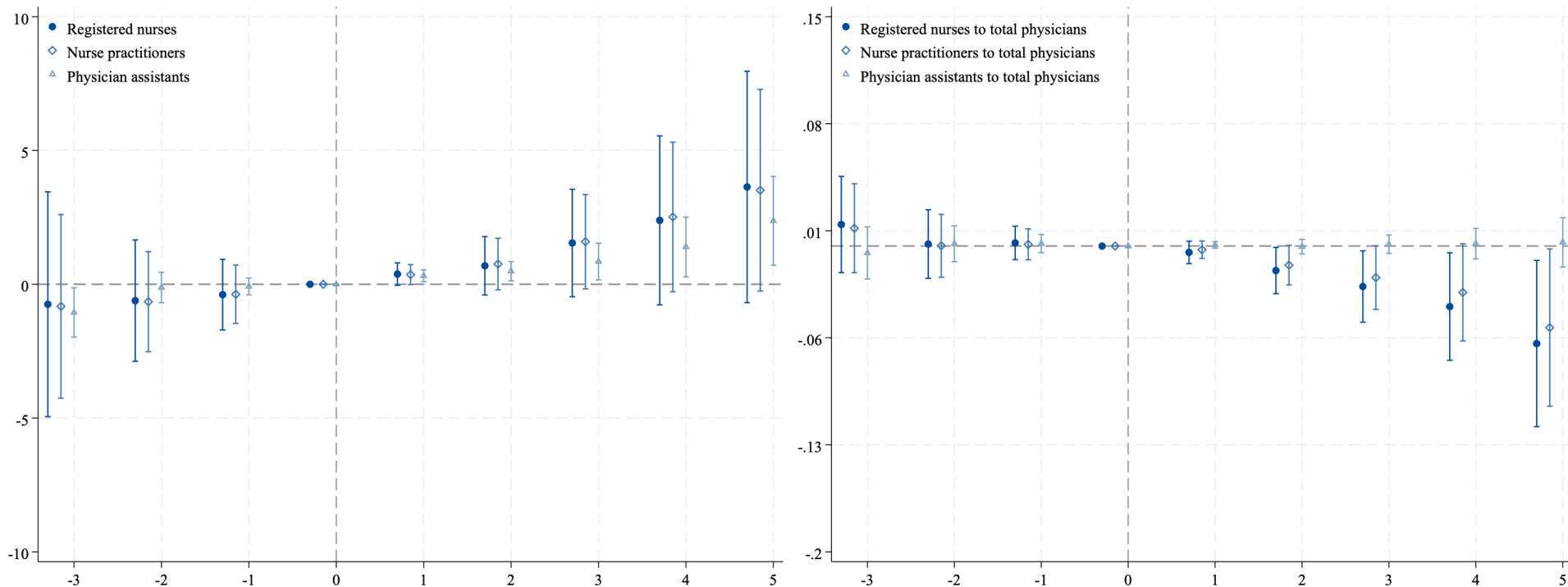
Potential Mechanism: the Financial Impact of HIT

- Is it worth it financially? We use proxies for physician/hospital's performance:
- [per physician] avg Medicare-allowed amounts (incl. Medicare payments, patient deductibles and coinsurance, and 3rd-party contributions), avg # of Medicare services
- [hospital] average hospital profit, revenue, and cost within each county



HIT and Physician Extenders: rates and ratios

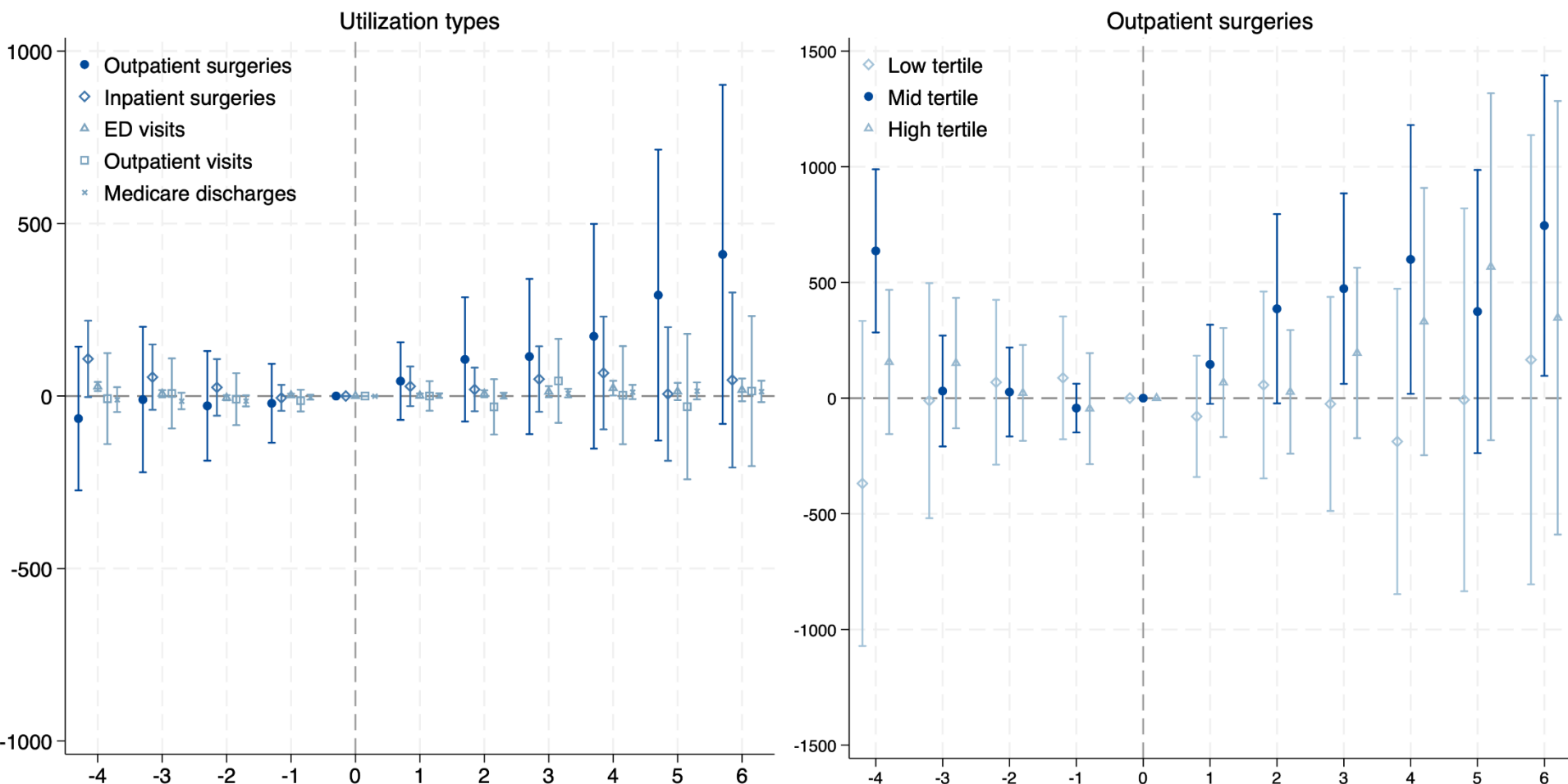
- PA-to-physician ratio remains similar; nurse-to-physician ratio declined, consistent w HIT automated routine charting tasks, freeing nurses to more direct patient care



Heterogeneity by utilization: doc supply & care vol.

- Knowing that \$ goes up. Given the interplay between **provider capacity** and **service demand**, how about **patient care utilization** post treatment?
- Specifically, whether **care utilization** responses to HIT diffusion vary by **pre-period local care utilization levels**?
- Tested across all utilization measures available, note that:
 - **Outpatient elective surgeries** has the **highest profit margins**, thus are where we expect highest increases in docs/vols (extensive, intensive margin)
 - Inpatient surgery volume may not respond, but more HB docs are still good
 - Others: ED visits, outpatient visits and Medicare inpatient discharges
 - Pre-period utilization matters: competition, disease mix, growth potential

Heterogeneity by utilization (pre-T p.c. Medicare tertile)



Separate tertiles based on pre-treatment care utilization, measured by standardized, risk-adjusted per capita Medicare costs in 2006

Robustness checks

- Alternative subsamples

subsample

- Counties with ≤ 5 distinct adoption rates; those never had zero HB physicians
- Counties not with extreme population sizes (exclude top/bottom 10%)
- Balanced panel with ≥ 4 periods before and after treatment
- Leave-one-out analysis: not driven by uncertain outlier specialties
- Matched DiD (top 10 counties, outside own census region)

drop-1-out

match-DD

- Alternative model specifications

specifications

- Models without state-specific time trends; without county level controls
- Models without either state-specific time trends or county level controls

- Alternative staggered DiD estimators or specifications

estimators

- Borusyal et al., 2024; Callaway & Sant'Anna, 2021; Sun & Abraham, 2021
- Continuous DiD; count data as DV (# rather than rates)

cont. DD

count DV

- Alternative HIT measures

measures

- CPOE or physician documentation (PD), another advanced HIT
- CPOE or PD or CDR (clinical data repository), not much effect with basic HIT

Conclusion & Discussion

- We find that aggregate HB physician density (county-level) **increases** after rapid adoption of advanced HIT, with effects concentrated among
 - Medical & surgical specialties: tied with data & scheduling
 - Early-career physicians: age ≤ 45 , more movable and tech savvy
 - Health professional shortage areas (HPSA): places w excess demand to care **limitations**
- Mechanism: treated counties showed better financial performance
 - Higher (per physician) avg Medicare allowed payments and services provided
 - Increased hospital profits, and greater outpatient surgical volumes
- Policy and managerial implications
 - Strategic HIT investment can attract (/retain) physicians to underserved areas
 - HIT (automation) appears to complement physician labor rather than substitute
 - Health tech diffusion may help reduce geographic disparity in care access

Health IT Diffusion and Physician Density



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Preview of Methods and Results

- Methods:
 - Event-study approach à la East et al. (AER 2023), leveraging the sharp increase in adoption rates to separate sample into treatment and control groups
 - Here: use sharp increase in county-level advanced HIT adoption rate
 - Compare HB physicians per 100k in counties with rapid diffusion vs controls
 - Use De Chaisemartin & d'Haultfoeuille (2024) as our main DiD estimator
 - Robust to other DiD estimators, continuous D, count data Y, other specifications
- Main results:
 - 11.1% increase in county-level HB physicians per 100k post HIT adoption
 - Effects primarily driven by medical and surgical specialties
 - Stronger effects among early-career docs and in areas with doc shortages
 - Physicians (hospitals) in treated counties received higher payments (profits)
 - Increased outpatient surgeries in counties w/ moderate pre-period volumes

Literature and Contribution

- **The impact of Health IT:** new evidence on how HIT automation affect aggregate physician labor density (high-skilled labor), and how the effects vary by specialty, career stage, shortage area status, and care utilization.

(Miller & Tucker, 2011; Agha, 2014; Dranove et al., 2014; McCullough and Snir, 2010; McCullough et al., 2016; Freedman et al., 2018; Lu et al., 2018; Greenwood et al., 2019; Wang, 2021; Wang & Bloch, 2023; Lin & Olson, 2024; Glenn, 2026)

- **Physician labor supply:** new technological angle that combines market-driven incentives and policy-led changes in the past decades, and study how digital technologies affect physicians, hospitals, and care provisions.

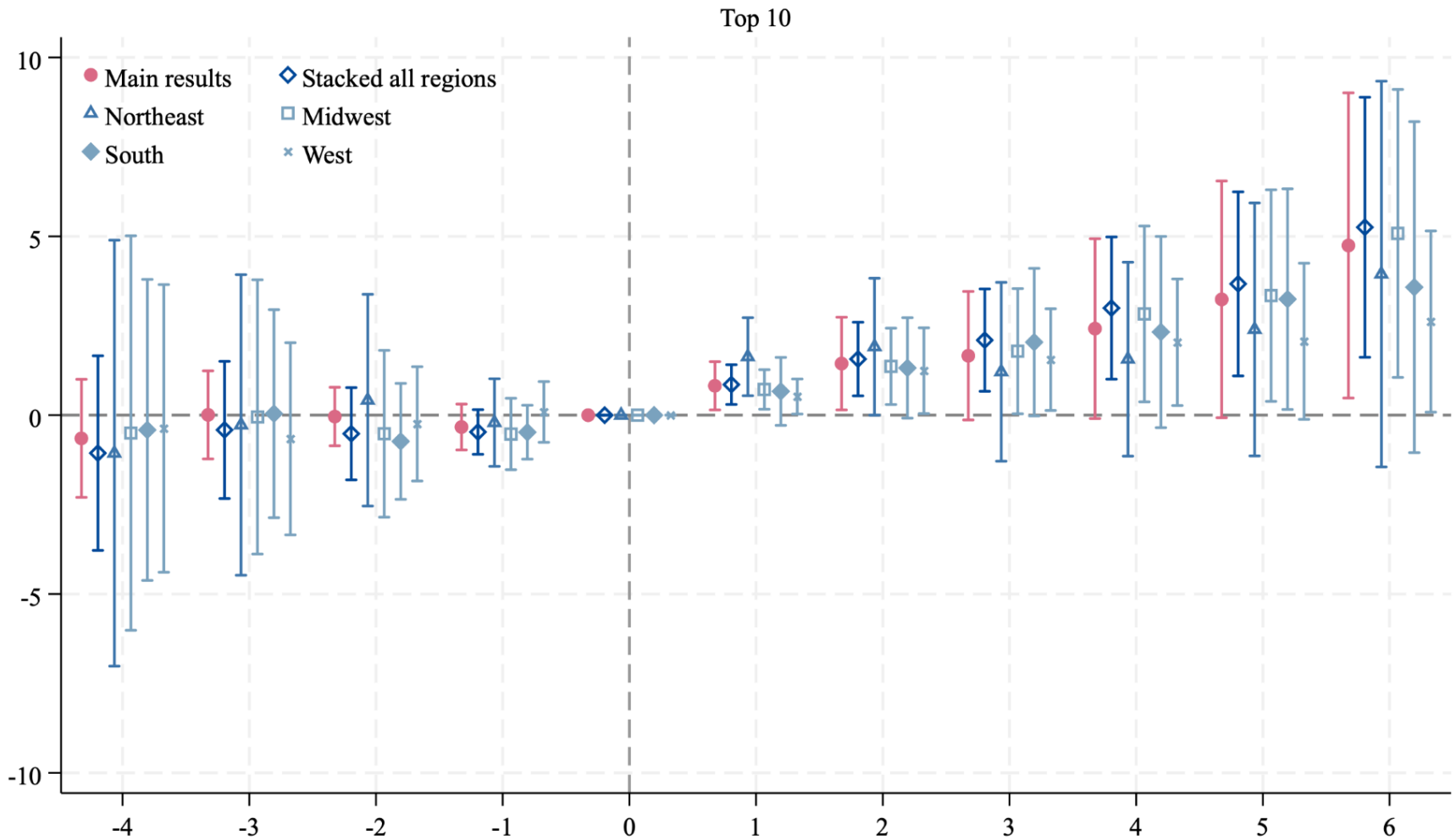
(Baker & Royalty, 2000; Garthwaite, 2012; Buchmueller et al., 2016; Falcettoni, 2018; Kulka & McWeeny, 2018; Neprash et al., 2018; Huh, 2021; Curto & Bhole, 2022; Huh & Lin 2024; Khoury et al., 2024)

- **Policy implications** and relevant insights for managerial practices. HIT investment can attract physicians, expand access to care (reduce disparity in access), while sustaining providers' financial health.

Summary statistics in pre-treatment period

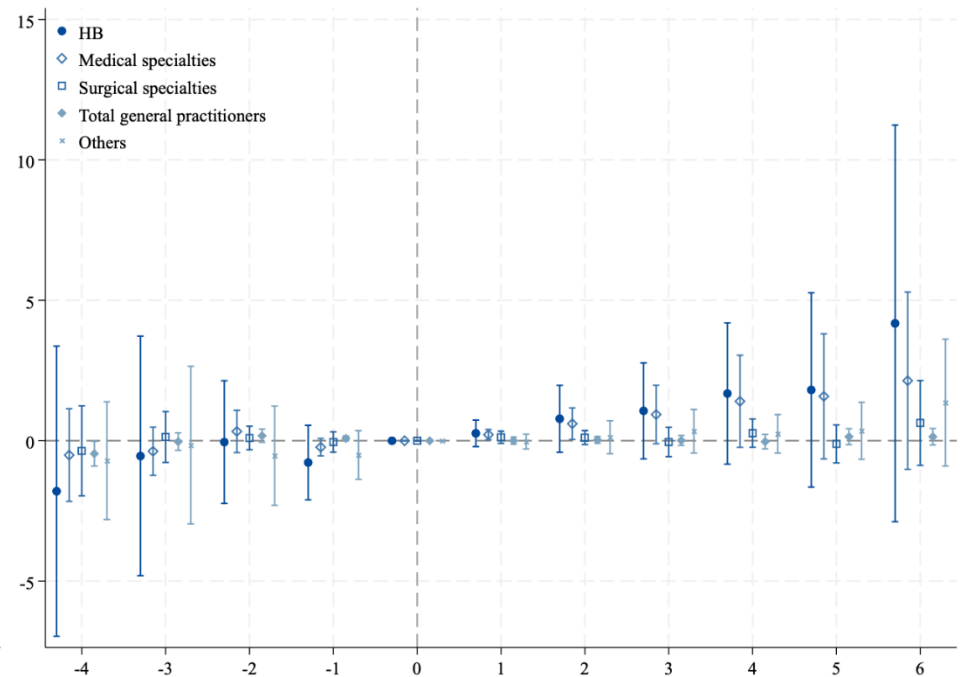
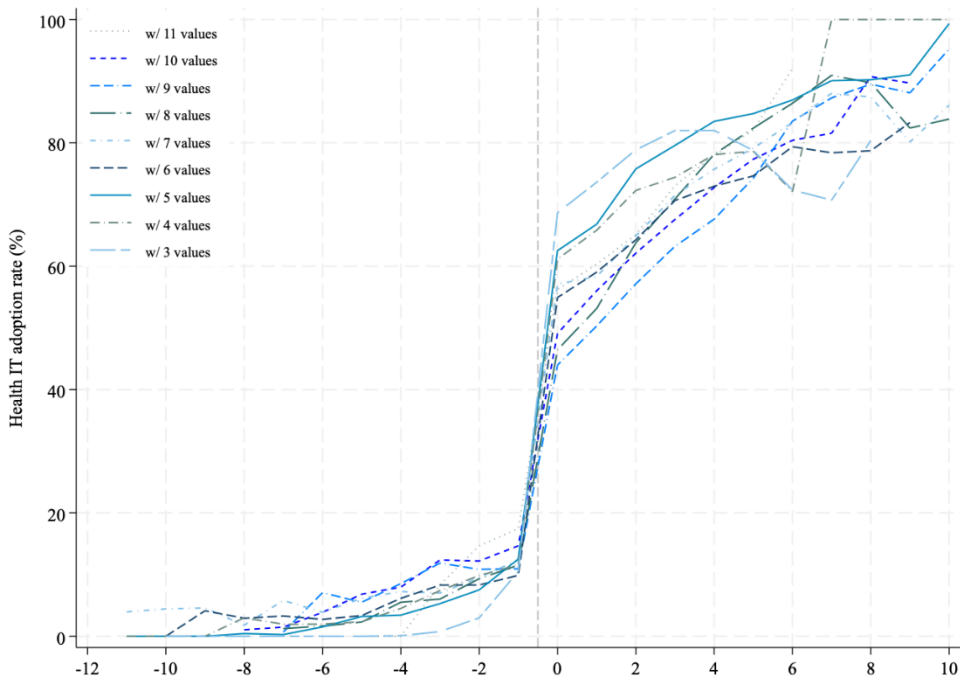
	Overall	Never-treated	Ever-treated
HB physician counts per 100k population	17.2 (37.8)	13.0 (17.0)	17.7 (39.6)
Health IT adoption rate (%)	1.46 (5.39)	0 (0)	1.64 (5.69)
% 65+ population	15.7 (4.07)	17.2 (4.35)	15.5 (4.00)
Poverty rate (%)	15.9 (5.95)	16.6 (6.10)	15.8 (5.92)
Income per capita	32,356 (8,325)	34,872 (10,304)	32,046 (7,996)
Medicare Advantage penetration rate (%)	4.06 (7.73)	2.87 (5.58)	4.21 (7.95)
% Medicaid eligibles	20.3 (8.35)	21.1 (9.10)	20.2 (8.25)
% above college degree	12.6 (5.7)	11.4 (5.1)	12.8 (5.7)
Hospital HHI	0.80 (0.27)	0.92 (0.15)	0.78 (0.27)
Population density	239 (1,722)	107 (520)	256 (1,815)
# counties	2,394	265	2,129

Matched-DiD: overall stacked & by census region

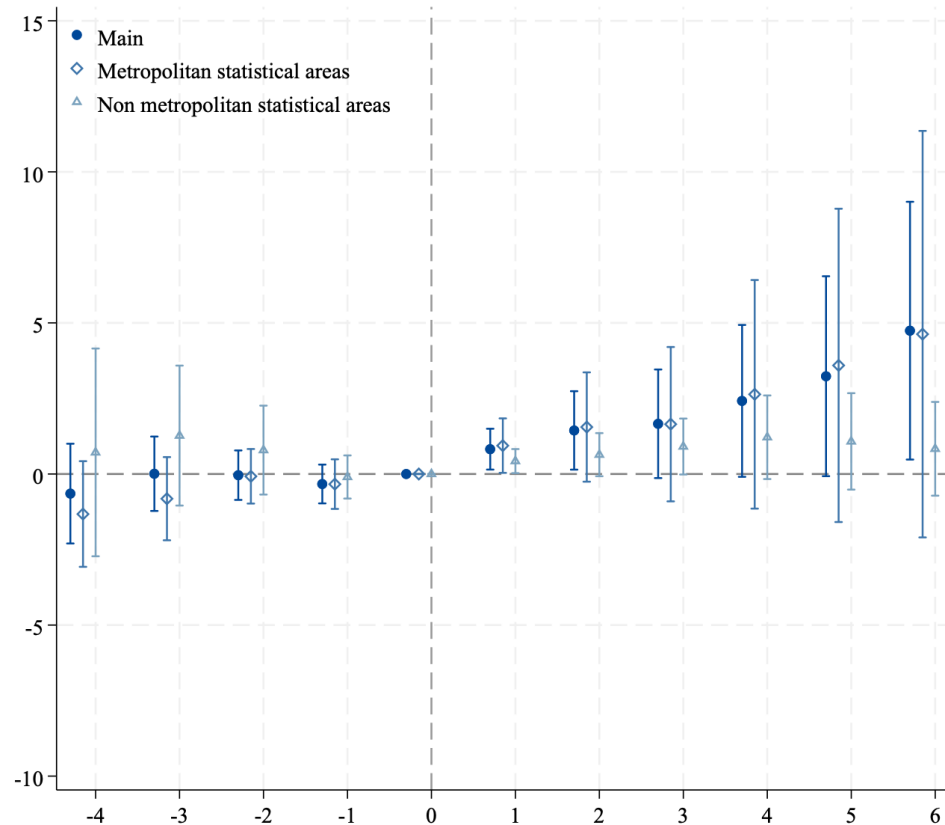
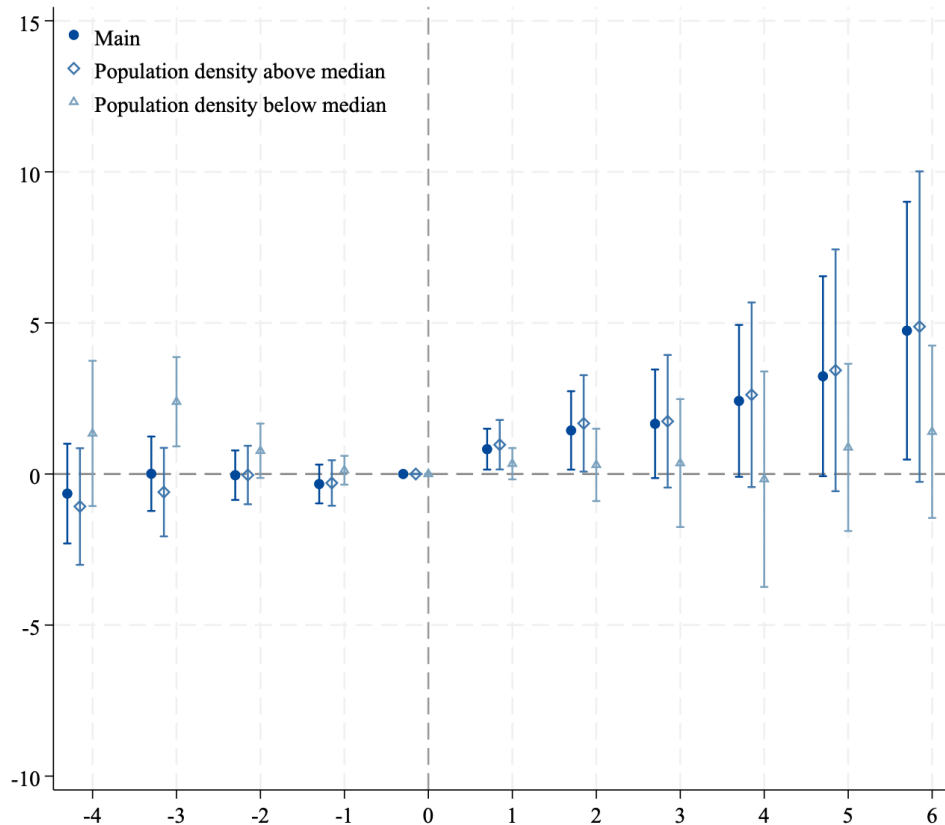


method à la Deryugina et al. (2018 AEJ) and identify the top 10 control counties for each ever-treated county, based on the following county characteristics in 2000-2004: HB physician density, annual % change in HB physician density, Medicare advantage penetration rate, % 65+ population, income per capita, and unemployment rate. We then further restrict the control counties to be far enough (≥ 300 miles and in a different census region from the focal county).

Hospital Referral Region (HRR) level analysis

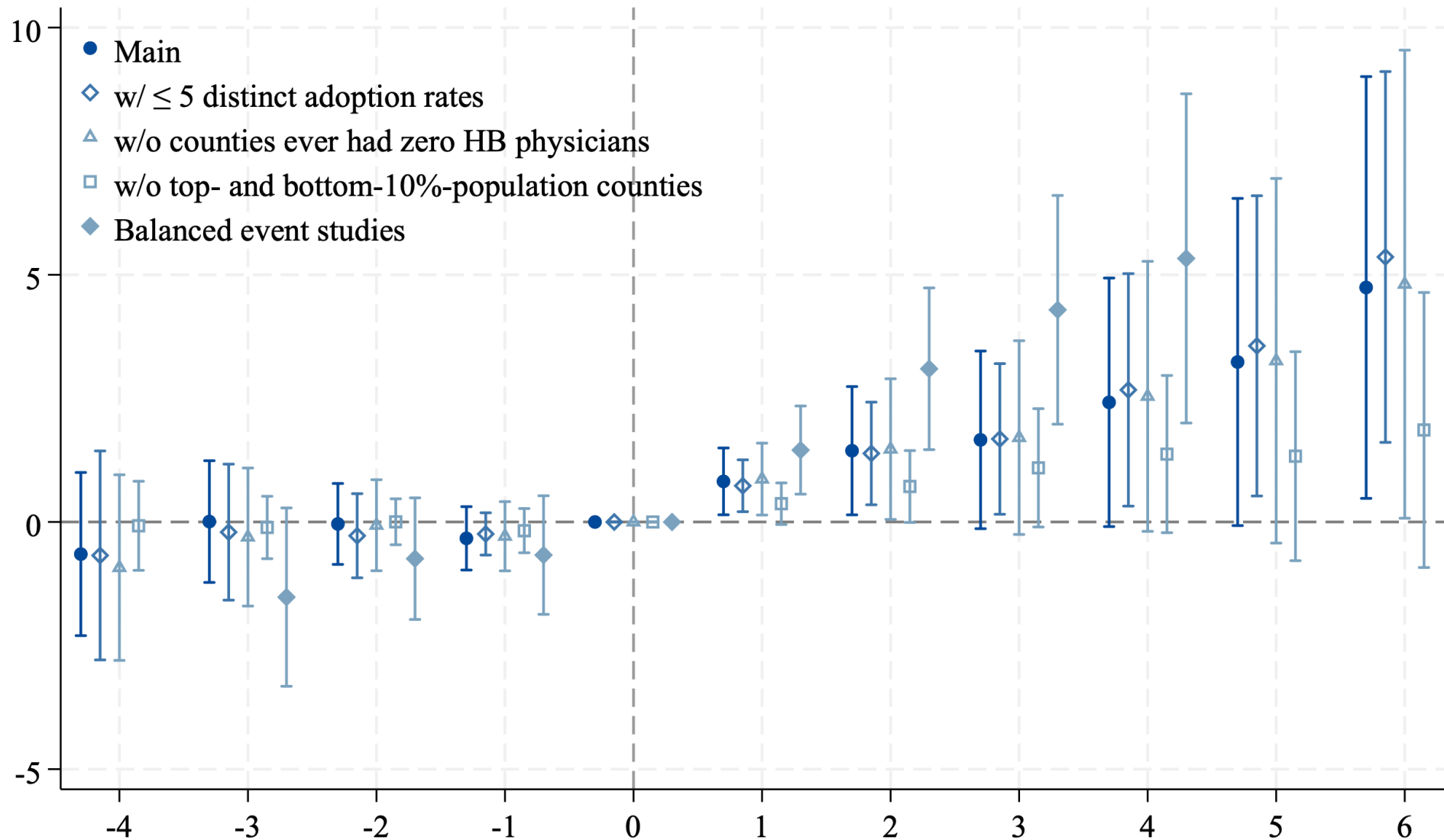


Heterogeneity by Rural/Urban (by pop density, MSA)

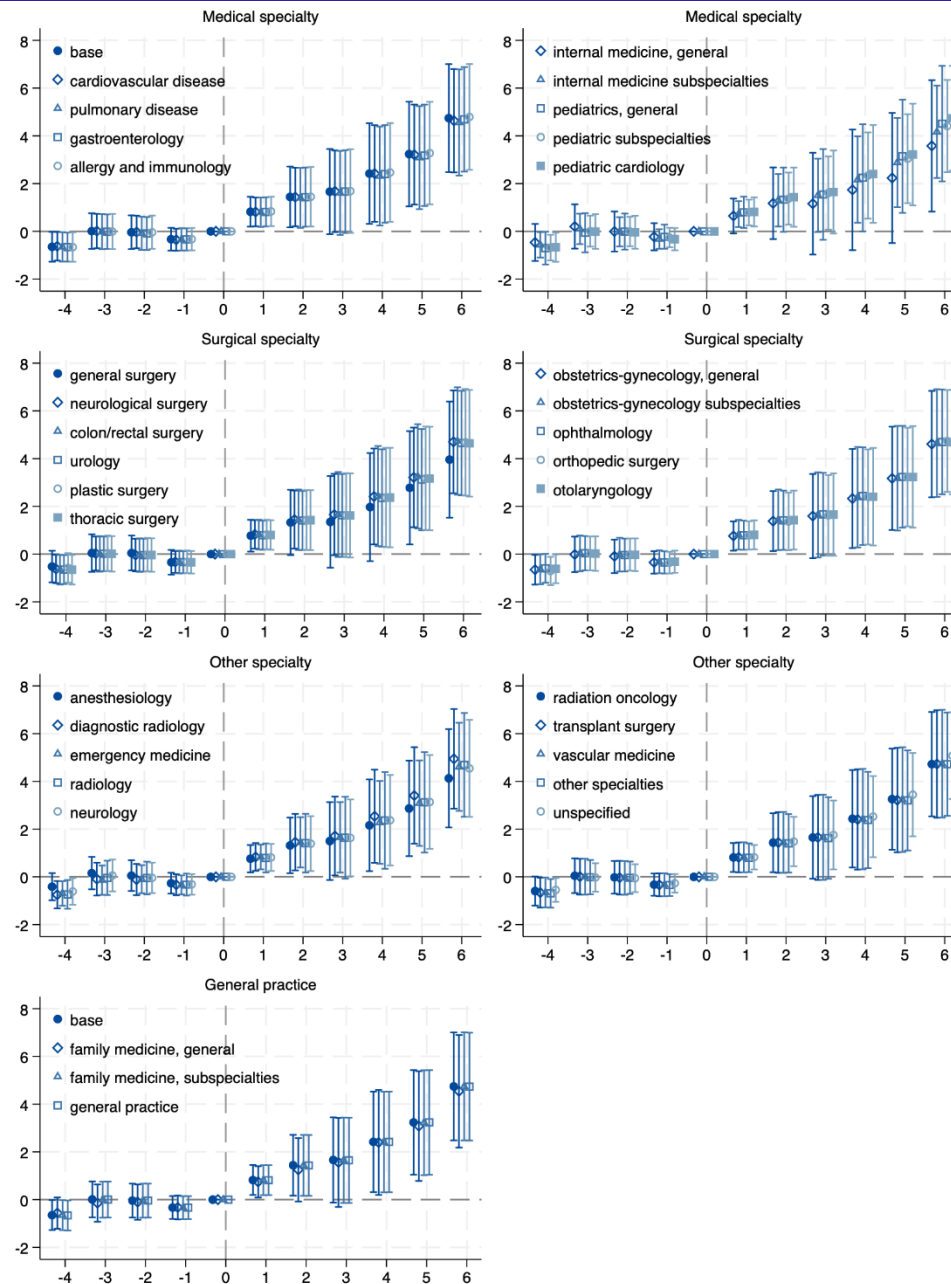


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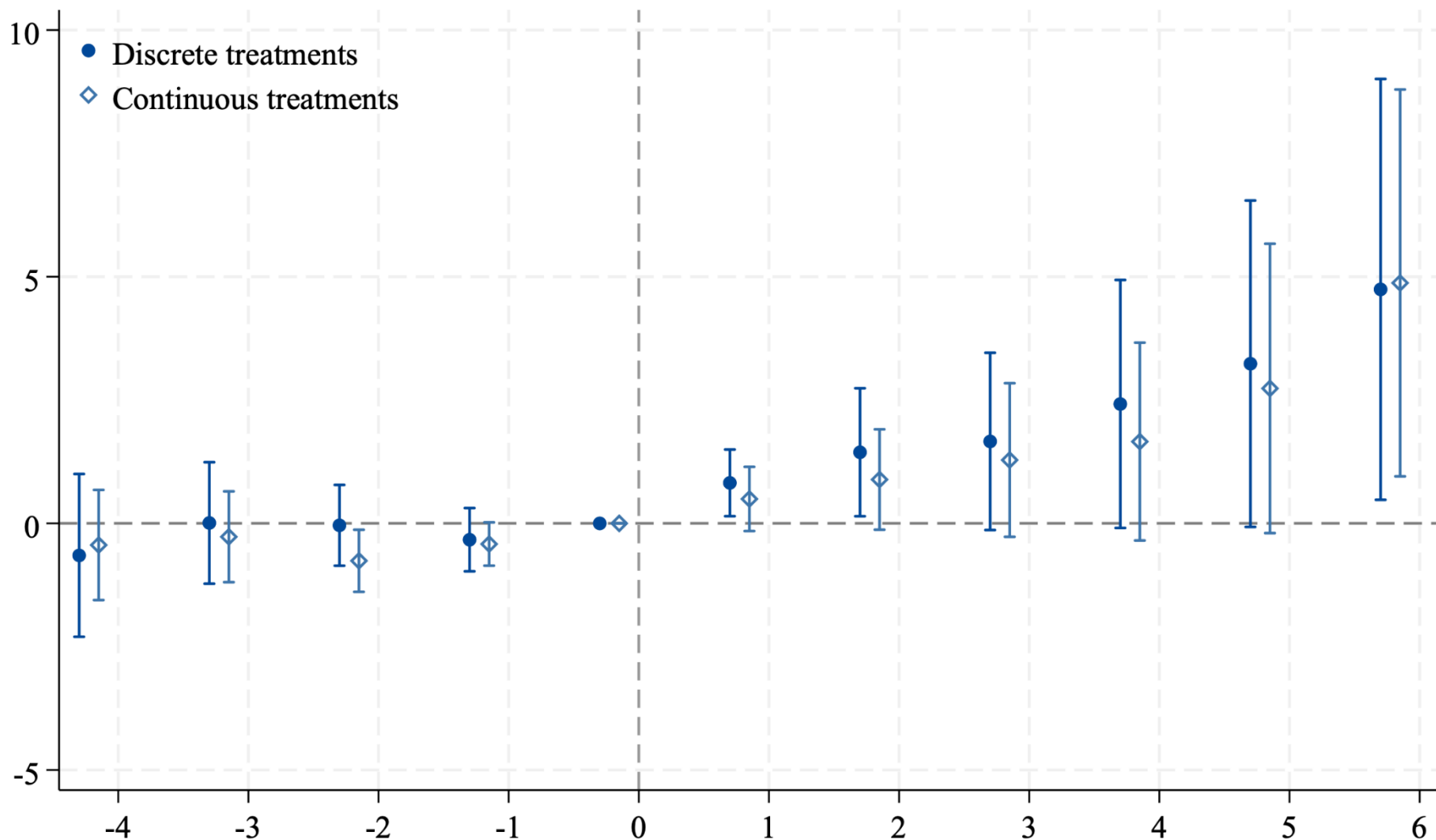
Robustness check: subsample sensitivity checks



Leave-one-out analysis by specialty

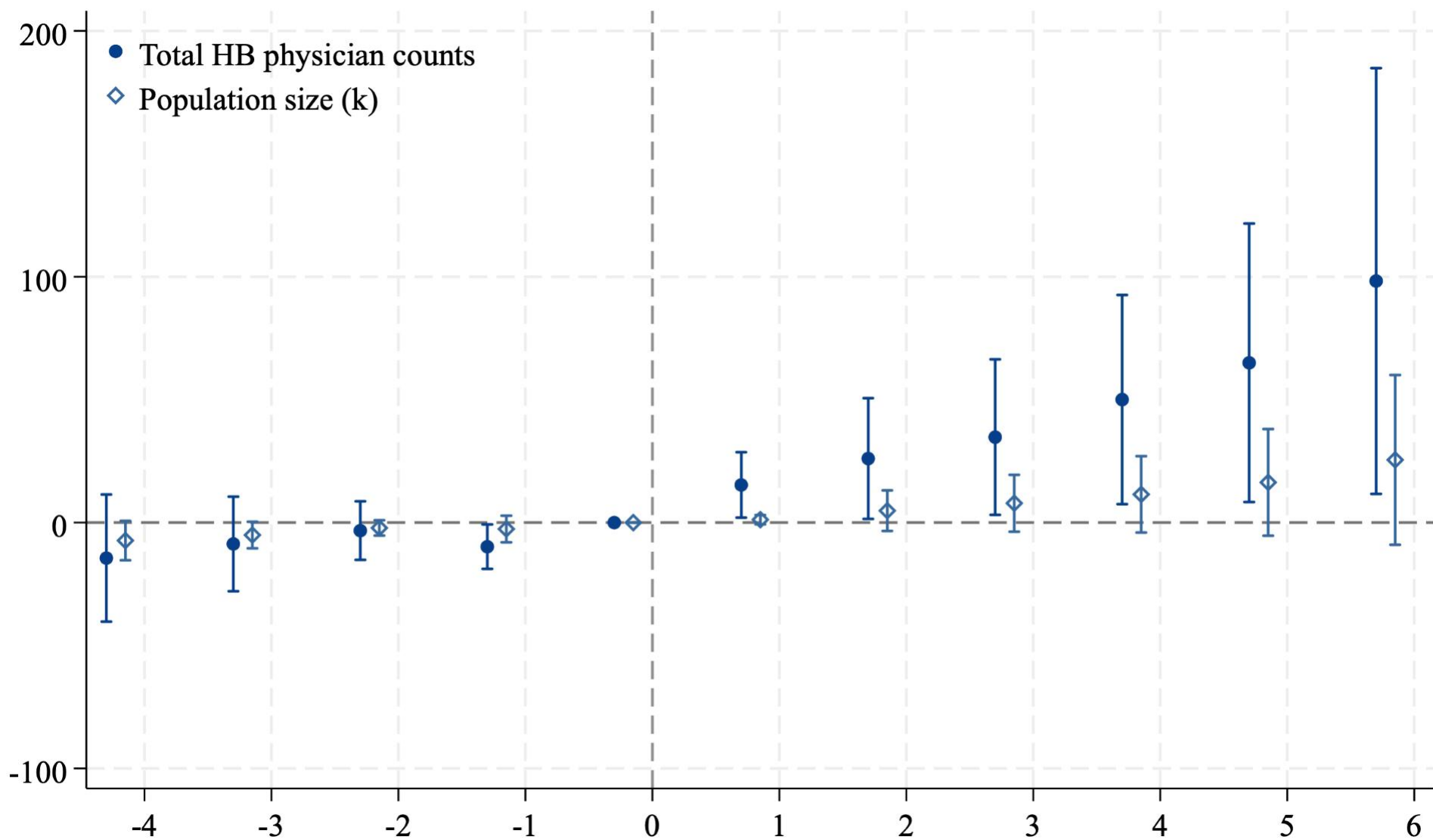


Discrete vs continuous HIT adoption treatment



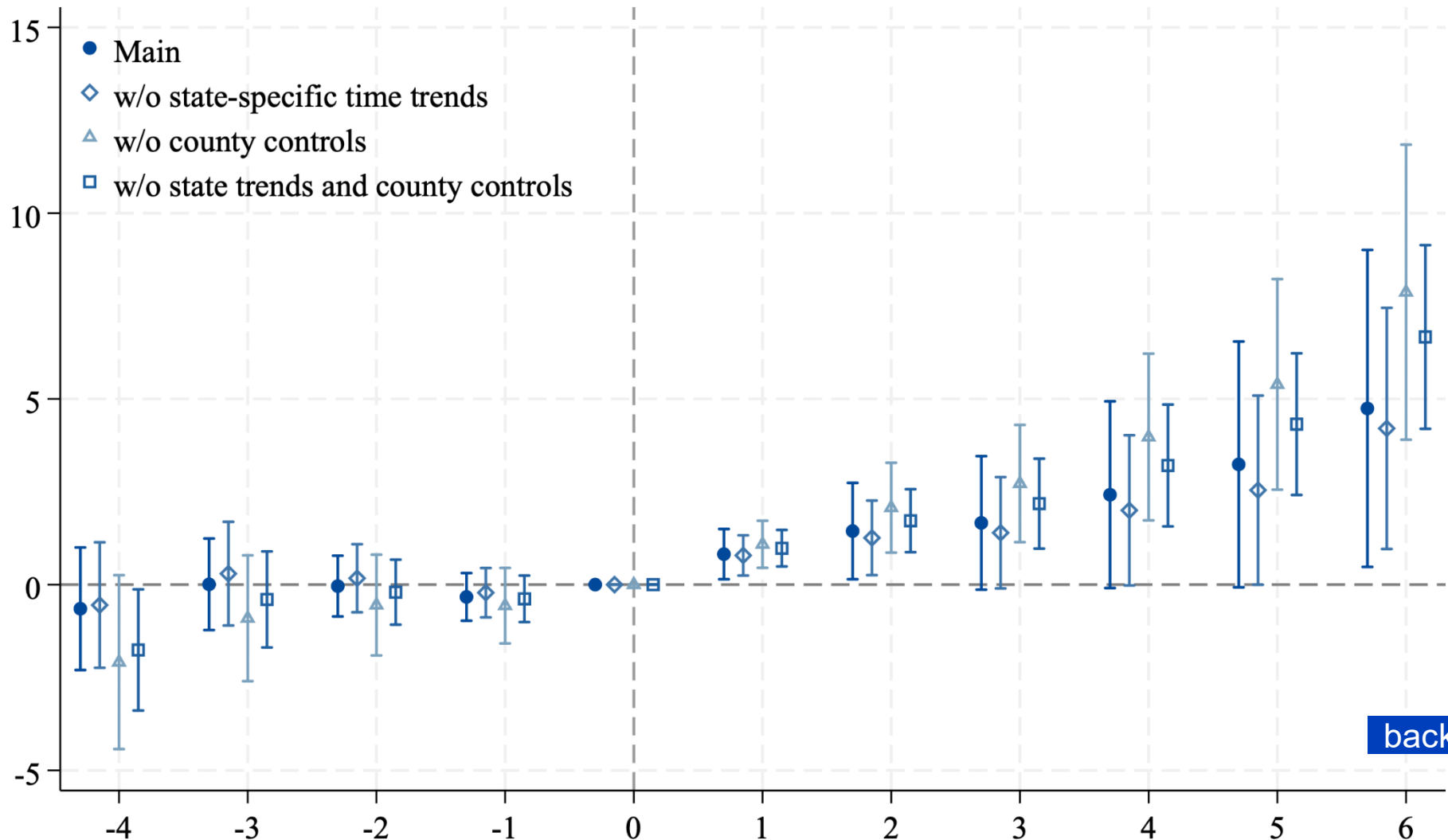
The estimated ATE from a continuous DiD estimation approach is 2.72** (1.35), compared to our main estimates under discrete DiD of 2.16** (1.04).

Count data DVs for main results (num./denom.)



Robustness check: specification checks

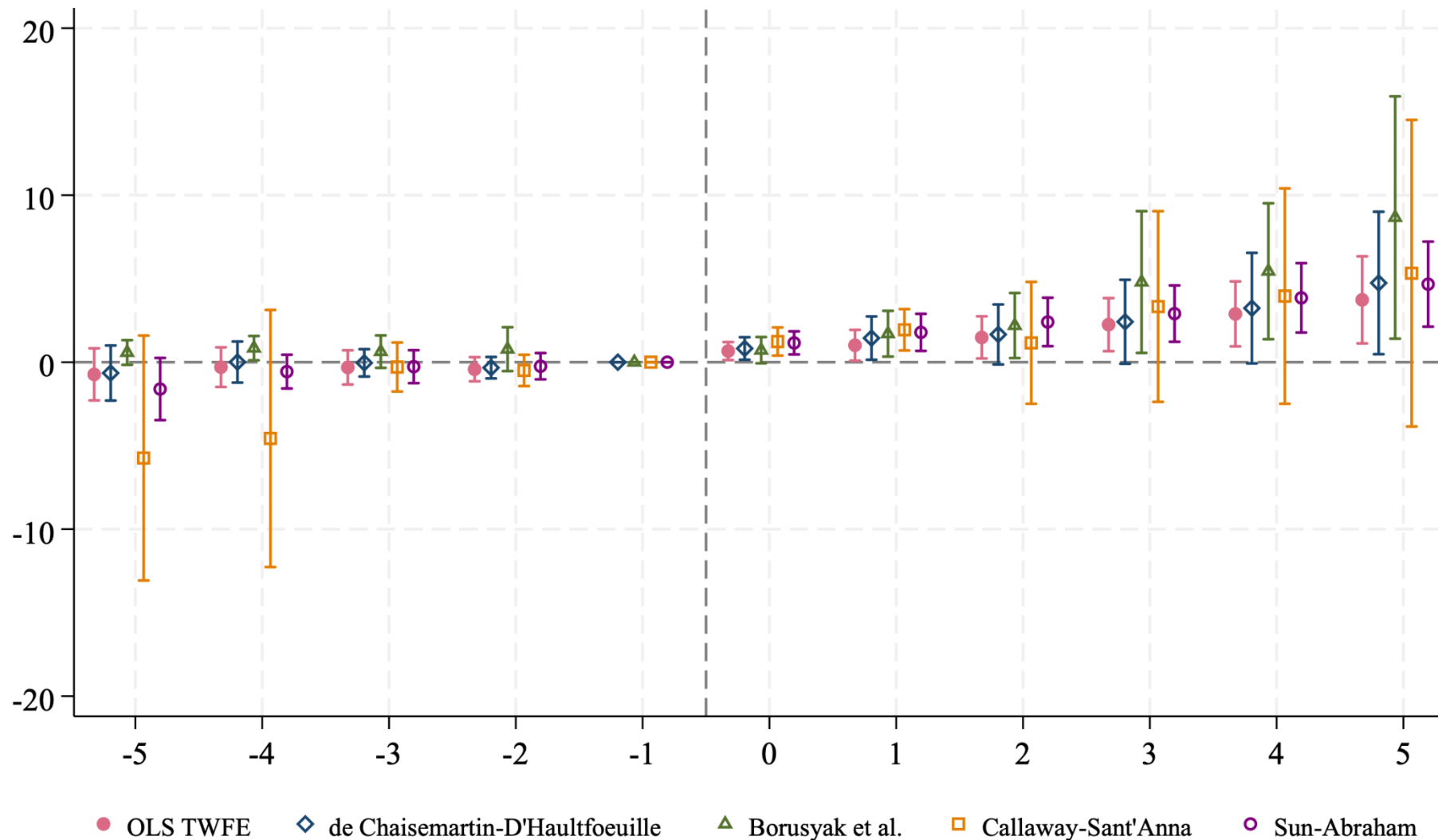
- Note the main specification always include state-specific time trends and a set of county-level observables in 2005 each times a linear time trend.



Alternative DiD estimators for the main outcome

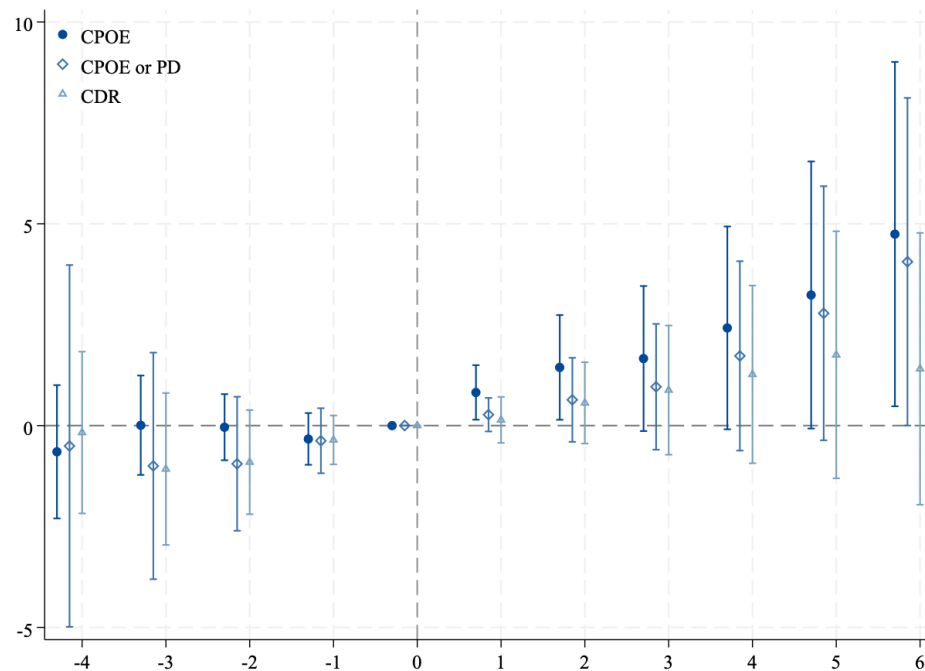
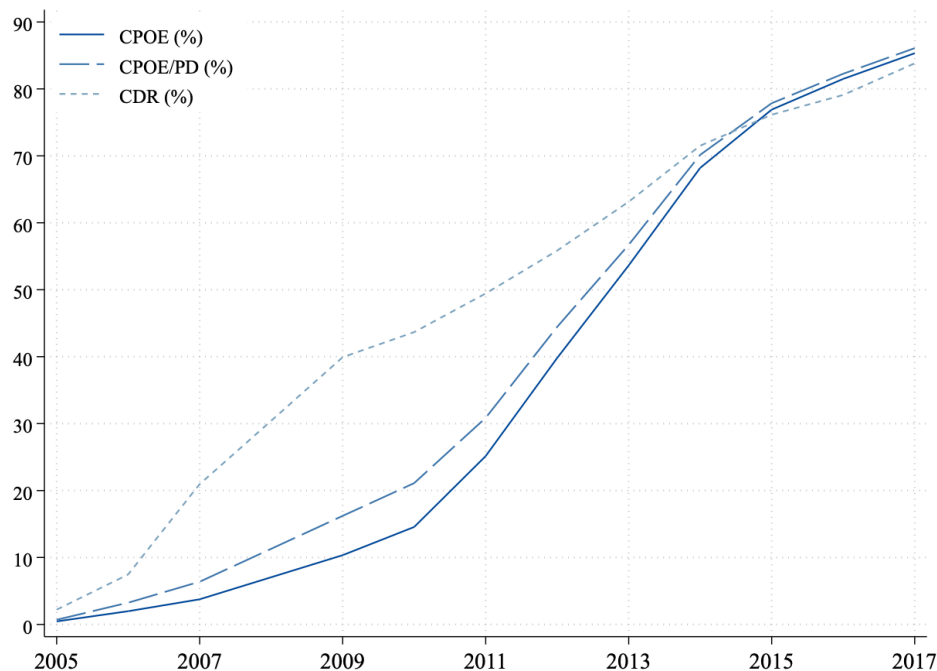
- All estimators use $t=-1$ as the reference except Borusyak et al., which uses the earliest period as the baseline period.

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Alternative HIT measures: CPOE/PD adoption

- Alt. HIT adoption measures: county-level sharp adoption of either CPOE or physician documentation (PD) – another advanced HIT component (noted in Dranove et al., 2014) but not core for HITECH act MU criteria.
 - Clinical Data Repository (CDR), in contrast, is a basic HIT w/ limited functions



Limitations and Future Work

- Analyses based on *aggregate* data
 - ☺ mitigate concerns about patient sorting (limited sorting at county level)
 - ☹ limit our ability to disentangle changes at the physician level (e.g., senior to junior doc replacements, PCP vs specialists, roles, interoperability, ...)
- Focus on *digital* tech but not fully capture *physical* med tech (data limit)
 - comprehensive controls mitigate concerns that other tech drive the results
- Not directly link with downstream *health outcomes*: HIT or docs?
 - HIT is one element within broader, team-based care delivery process
 - We demonstrate there can be direct and indirect effects via physician supply